GreenSignal

Introduction

GreenSignal is an application project for a Leasing Management System. This system aims to help leasing companies manage customer financing applications. Meanwhile, customers can submit leasing requests, and internal officers can review, survey, approve, or reject applications efficiently.

Submitted Contents

The accepted answer file types are .png and .drawio, and all submissions must follow the folder structure below:

- Use case
- Activity Diagram
- ERD
- Wireframe

After After organizing your files according to the structure above, compress them into a .zip file using the naming format: **SD_[PC_NO].zip.**
Submission that do not meet these criteria will not be evaluated.

Description of project and tasks

While developing the test project, please make sure the deliverables conform to the basic guidelines given:
• There should be consistency in using the provided style guide throughout development.
• The use of valid and proper naming conventions is expected in all material submitted.
• Time management is critical to the success of any project and so it is expected of all deliverables to be complete and operational upon delivery.

Application Requirements

1. User Roles

There are 3 main user roles in this application, they are Customer, Surveyor and Back Office Officer. The user must log in to the application before using it. Customers and Surveyors will be using the application on mobile. Meanwhile, the Back Office Officer will be using the application on the desktop. Some pages might have different pages or actions depending on the user who signs in.

Customers are users who want to apply for leasing or financing of an item or vehicle. They can submit applications, upload documents, and check application progress.

Surveyors are responsible for conducting field verification and submitting survey results and recommendations.

Back Office Officers are responsible for reviewing the flow of application. They could decline the application back to customer if the submitted application need revision, or assign the application to surveyor. After surveyor submit the survey result, they could review it again to approve or reject the application based on the analysis.

2. Submit Leasing Application Page

• This page will be used by the Customer to submit a leasing application or edit their application if their application need revision.

• The data required to be submitted are:

Field	Description
Customer Information	All input based on ID such as: name, ID number, phone, address, Date of Birth, and occupation.
Leasing Object	Leasing type available only for Motorcycle or Car, also the chosen type of the vehicle will determine the list of brand, model, year, and price.
Financing Details	There will be input for: ● Down Payment (the portion of the total price that you pay **upfront** before the financing (the loan)

	starts) ● Tenor (the **length of time in months** you have to pay back the remaining balance).
Income Information	Monthly Income and assets data.
Supporting Document	Upload the ID Card, proof of income, also the agreement.

- After submitting the data, the page will generate an installment plan with its interest, the interest will be fixed and if between 0 - 12 months the interest will be 0.5% then 12-36 will be 2% and 36-60 months it becomes 5% and the rest with maximum 10 years will be 6.5%. After reviewing it, customers could submit the leasing application.
- The submitted application will be available for review by the Back Office Officer based on the submitted date.

3. View Application Page

- This page can be used by all roles.
- For Customers, this page shows all submitted applications with its status and could revise their application if required.
- For the Admin Officer, this page shows all incoming applications.
- For Surveyors, this page shows assigned applications.
- For Back Office Officers, this page shows applications from customer (pre survey) and surveyor (post survey).

4. Review Application Page

- This page is available only for Back Office Officers.
- They can check completeness of supporting documents and confirm the user's credentials such as address, name, and phone, so they could assign the surveyor based on the location.
- If incomplete, reason must be provided.

5. Conduct Survey Page

- This page is available only for Surveyors.
- The surveyor's goal is to verify the customer by filling survey results (such as confirming the data validation also could correct if there’s changes in customer information), uploading evidence like the truth about customers via photos, geo-location for the address, and maybe the survey recording. and gives recommendations when submitting the survey result.

6. Final Decision Page

- This page is available only for Back Office Officers.
- Officers can approve or reject applications based on the application review, survey review, and also analyze the capability of the customer to finish the loan, usually by calculating Debt Service Ratio (DSR) with formula below:

DSR = **Total Monthly Debt / Monthly Net Income * 100%**
Total Monthly Debt: This includes your *new* installment plan plus any other existing debts in this company.
Monthly Net Income: Your "Take Home Pay" after taxes.

- The Back Office Officer could set the auto rejection threshold for the maximum DSR
- Rejection must include reason, but if it is because of the auto rejection the reason will be automatically registered as “DSR values are not in the acceptable threshold”.

Deliverables

1. Use Case Diagram

Use Case Diagram based on the given requirements. Use Case Diagram is the primary form of system/software requirements for a new software program underdeveloped. Use cases specify the expected behavior (what), and not the exact method of making it happen (how) (Visual Paradigm, 2024). The following Use Case can be used as the reference standard of your deliverables.

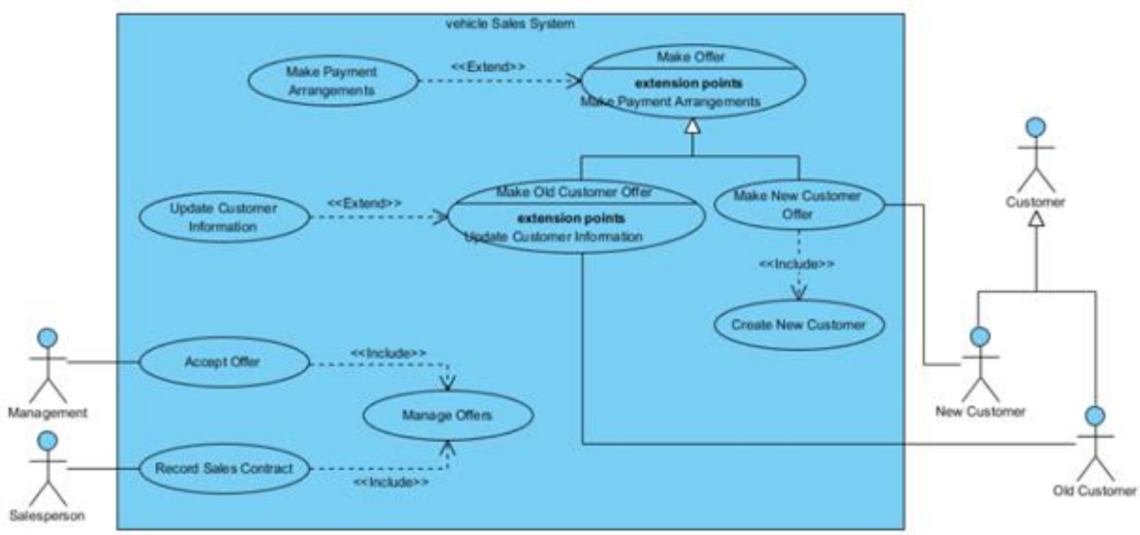

Figure 1. An Example of Use Case Diagram (Visual Paradigm, 2024)

2. Activity Diagram

for each case you list in the Use Case Diagram. Activity diagram is another important behavioral diagram in UML diagram to describe dynamic aspects of the system. Activity diagram is an advanced version of flowchart that models the flow from one activity to another activity (Visual Paradigm, 2024). The following Activity Diagram can be used as a reference standard for your deliverables.

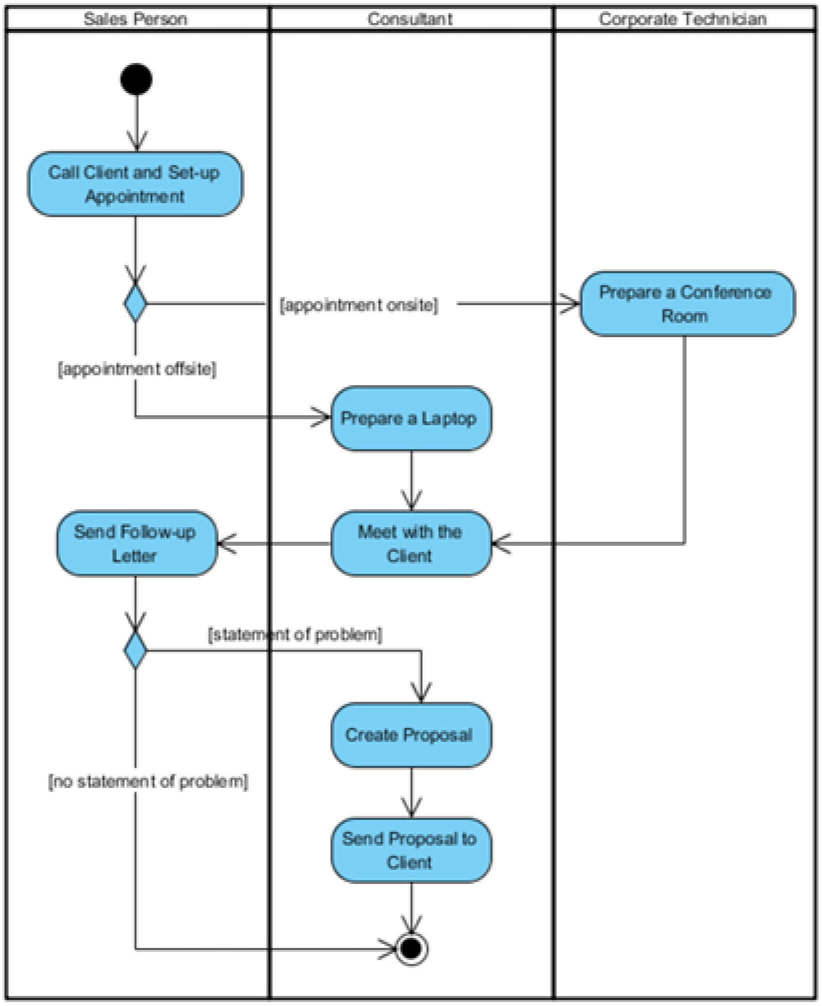

Figure 2. AN Example of Activity Diagram (Visual Paradigm, 2024)

3. Entity Relationship Diagram (ERD)

for the system based on the given requirements. ERD is a type of structural diagram for use in database design. An ERD contains different symbols and connectors that visualize two important pieces of information: The major entities within the system scope, and the inter-relationships among these entities (Visual Paradigm, 2024). The diagram will need to provide Entities (name, fields name and type), Relationship and Cardinalities. The following ERD can be used as the reference standard of your deliverables.

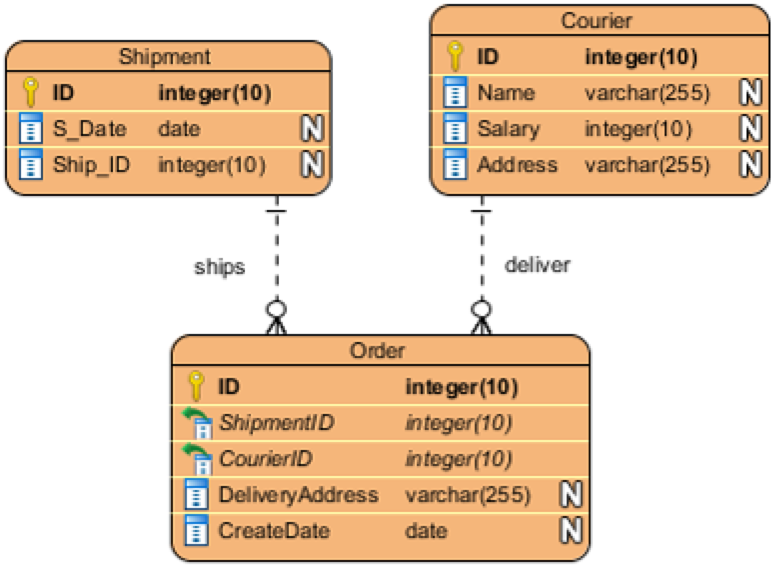

Figure 3. An Example of Entity Relationship Diagram (Visual Paradigm, 2024)

4. Wireframe / UI Design

Wireframe / UI Design for each page based on the given requirements. The following are some of the details required:

- A wireframe in digital design is a visual guide or page schematic that is stripped of typographic style, colors, graphics, and interactive elements, and represents a specific point in the design process. Its purpose is to show page-level layout ideas that depict functionality, behavior, and priority of content.
- All the required fields are to be included to make it functional.
- The use of a dialogue box as the container is recommended.
- Clearly labelled buttons and controls are required.
- Proper use of separators and indicators is necessary.
- Effective use of input fields (drop-down and list boxes) is essential.